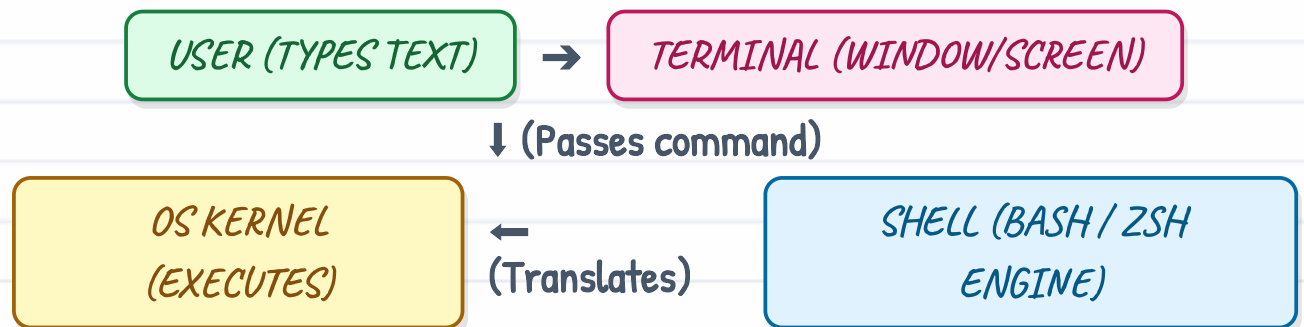


TOPIC 01: WHAT IS TERMINAL, CONSOLE, SHELL & BASH?

Definition:

Terminal ek **Text Input and Output Environment** hai jahan mouse ke bina, sirf text commands type karke hum computer se directly interact karte hain.



Terminology Clear Samjho:

- **Command Line (CLI):** Koi bhi textual interface jisme text commands daale jaate hain (General Windows centric concept).
- **Terminal:** CLI ka ek visual window program jo input leta hai aur output render karta hai (Mac/Linux centric).
- **Console:** Physical ya low-level system screen jahan computer ke core system messages aate hain.
- **Shell:** Terminal ke andar chalne wala actual **Interpreter Program** jo commands ko computer ki language mein convert karta hai.
- **Bash:** "Bourne Again Shell" — Linux aur purane macOS ka sabse popular standard shell.
- **Z-Shell (zsh):** Modern macOS ka naya default shell (fast, advanced auto-suggestions & themes).

TOPIC 02: BASICS (ESSENTIAL COMMANDS)

3 Core Commands:

Command	Full Form	Simple Hinglish Meaning	Example
pwd	Print Working Directory	"Main abhi computer ke kis folder mein khada hoon?"	/Users/shivam/project
ls	List files	Current folder mein kaun-kaun si files/folders hain unki list dikhata hai.	index.html app.js
clear	Clear Screen	Terminal screen ke saare purane text ko clean/saaf kar deta hai.	Shortcut: Ctrl + L

TOPIC 03: NAVIGATION COMMAND (CD)

cd = Change Directory (Folder badalna):

`$ cd backend` # "backend" folder ke andar chala jayega `$ cd ..` # 1 step piche (Parent directory mein aana) `$ cd ~` # Direct Home directory mein jump karna `$ cd /` # Direct Root (base) directory mein jana



Exam Tip: `cd ..` mein double dots (..) hamesha parent folder ko represent karte hain!

TOPIC 04: PATH (RELATIVE VS ABSOLUTE)

Types of Path:

Path Type	Definition & Kaam	Example
Absolute Path	Root (/) se shuru hone wala complete full address. Har jagah se same kaam karega.	/Users/shivam/Downloads/Javascript
Relative Path	Current location ke reference se path batata hai (jahan aap abhi ho).	./notes.txt ya ../images

Special Path Symbols:

- / → Root Directory: Computer ka sabse pehla / top-most folder.
- ~ → Home Directory: Current logged-in user ka personal space (e.g. C:\Users\shivam).
- . → Current Directory: Jahan abhi aap khade hain.
- .. → Parent Directory: Current folder se ek level upar.

TOPIC 05: MAKE DIRECTORY (MKDIR)

mkdir = Make Directory (Naya Folder Create Karna):

```
$ mkdir my_project # "my_project" naam ka single folder banega
$ mkdir folder1 folder2 # Ek sath multiple folders banane ke liye
$ mkdir -p parent/child # Nested folders ek sath banane ke liye (-p = parents)
```

TOPIC 06: FLAGS & MANUAL (MAN)

Flags Definition:

Flags wo extra characters hote hain jo hum command ke aage - (hyphen) lagakar pass karte hain taaki uska **default behavior modify** ho sake.

Manual Command (man): Kisi bhi command ka pura guide/manual padhne ke liye (e.g., `man ls`, `man mkdir`). Bahar aane ke liye `q` press karte hain.

Common Flags with ls:

- `ls -l` → **Long format:** File permissions, size, owner, date & time sab show karta hai.
- `ls -a` → **All files:** Hidden files bhi show karta hai jo dot (.) se start hoti hain jaise `.gitignore`, `.env`.
- `ls -la` / `ls -al` → **Combined:** Hidden files + Long format details dono ek sath dikhata hai.

TOPIC 07 & 08: TOUCH & DELETING FILES/FOLDERS

Topic 07: Touch Command:

Nayi empty files create karne ke liye with extension:

```
$ touch index.html # HTML file ban gayi $ touch app.js # JS file ban gayi $ touch abc.txt # Text file ban gayi
```

Note: Agar file pehle se maujood ho, toh touch file delete nahi karta balki uska **Access & Modification Timestamp** update kar deta hai.

Topic 08: Deleting Files & Folders:

Command	Use Case	Important Rule
rm <file>	Remove file (e.g. rm abc.txt)	Sirf files delete karta hai.
rmdir <folder>	Remove empty folder	Agar folder ke andar files hain toh error dega!
rm -rf <folder>	Remove any folder forcefully	-r = Recursive (files + subfolders), -f = Force (no asking).



Caution / Pro Tip: Terminal se delete hui files Recycle Bin / Trash mein nahi jaati! Wo permanently delete ho jaati hain, isliye `rm -rf` dhyan se use karein.